NCBI BLAST PRACTICE

BLASTN comparison of the human HBB reference transcript (NM_000518.5) with cDNA clone BC007075.1

Objective:

To compare the transcripts of human HBB RefSeq NM_000518.5 with human HBB cDNA clone BC007075.1 from NCBI and practice comparing nucleotide sequences using BLASTN, including understanding the alignment score, query coverage, percent identity, and E-value.

Tools:

- NCBI Gene and Nucleotide databases

- NCBI BLASTN

Results:

Comparison of all parameters between human HBB NM_000518.5 and with human HBB cDNA clone BC007075.1

PARAMETERS	HIT1	HIT2
Accession	HBB NM_000518.5	clone BC007075.1
Description	HBB, mRNA (refSeq)	HBB mRNA, CDNA clone MGC:14540
Scientific name	Homo sapiens	Homo sapiens
Max score	1160	1142
Total score	1160	1142
Query cover	100%	100%
E-value	0.0	0.0
Percent identity	100.00%	99.52%

Accession length	628bp	658bp

Accession: It is a unique identification number given to a sequence in the NCBI database. NM_000518.5 is the accession number of the HBB reference transcript and BC007075.1 human HBB cDNA clone used in this task.

Max score: Represents the highest alignment score for a single alignment segment between the query and the subject sequence.

Total score: It is the sum of score from all alignment.

Query Coverage: It means how much of query sequence is included in the alignment. A 100% query coverage means the entire sequence aligns with the subject sequence.

Sequence Identity: It is the number that describe how similar the query sequence is to target sequence means how many characters in each sequence are identical, higher the percent the identity is more significant to match.

E-value: Measures the quality of match, lower E- value indicate higher similarity.

Percent identity: It shows the proportion of nucleotides that are exactly identical between the query and the matched sequence. 100% indicate a perfect match.

Accession length: It refers to the total length of the subject sequence stored un the database.

Interpretation:

Hit 1 (NM_000518.5). This is the query matching itself, so it has 100% identity, 100% coverage and the maximum possible score (1160) with E-value of 0.0. It serves as the baseline for the comparison.

Hit 2 (BC007075.1). The clone covers 100% of the query with 99.52% identity and an E-value of 0.0, so the match is highly significant and the two sequences are highly similar and represent closely matching HBB transcript-related sequences.. Over a 628 bp alignment, 99.52% identity corresponds to roughly very few differing positions.

BC007075.1 has a score of 18 points lower than NM_000518.5 (1142 and 1160) shows few differences between the sequence.

BC007075.1 is 658 bp, about 30 bp longer than NM_000518.5 (628bp). Because the query is fully covered, the extra length lies outside the aligned region.

Conclusion:

In this task, I practiced how to obtain human HBB NM_000518.5 and human HBB cDNA clone BC007075.1 from NCBI, compare and the nucleotide sequence to BLASTN and interpret alignment score, query cover, percent identity, E-value, accession number.